Autonomous Infrastructure

Self-Adaptive Digital Twins for Dynamic Resource Management

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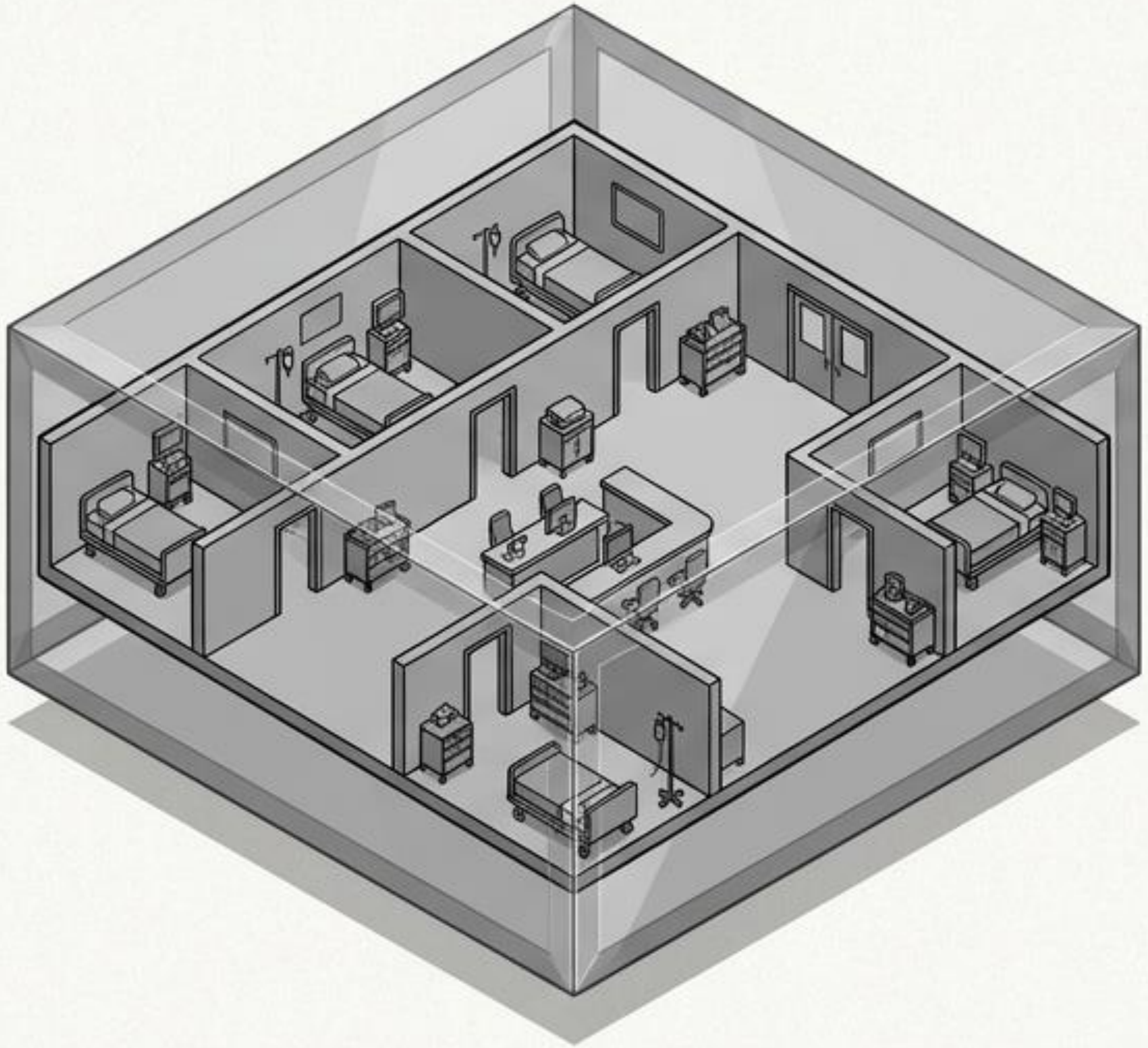
The Living Digital Twin

A self-adaptive architecture for dynamic resource management.

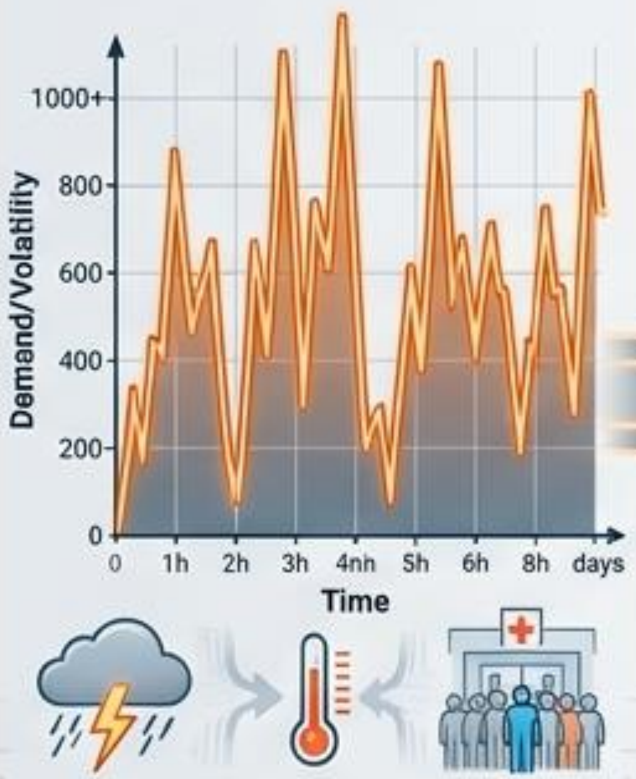
The Past: Static Systems



The Future: Living Digital Twins

The Core Problem: The Static Model Trap

Dynamic Reality



Unpredictable patient surges, extreme weather, and seasonal flu.

The Static Twin

Cannot adjust internal models or control policies as the real physical system changes.

Model Drift

Blindspots emerge when physical reality outpaces the rigid virtual model.

Manual Crisis Management

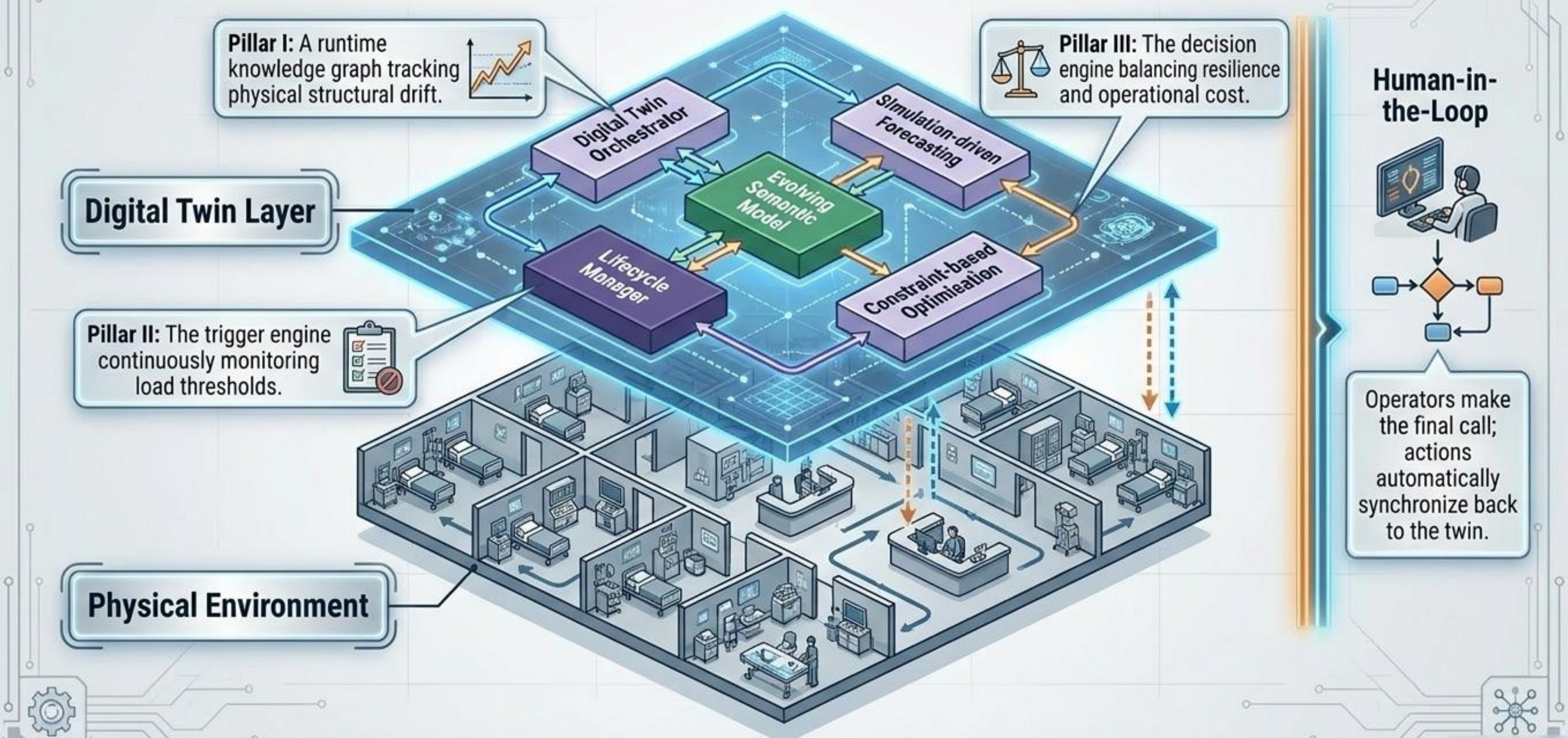
Operators are forced into makeshift solutions (e.g., placing patients in corridors), leading to massive inefficiency and systemic risk.

The Paradigm Shift

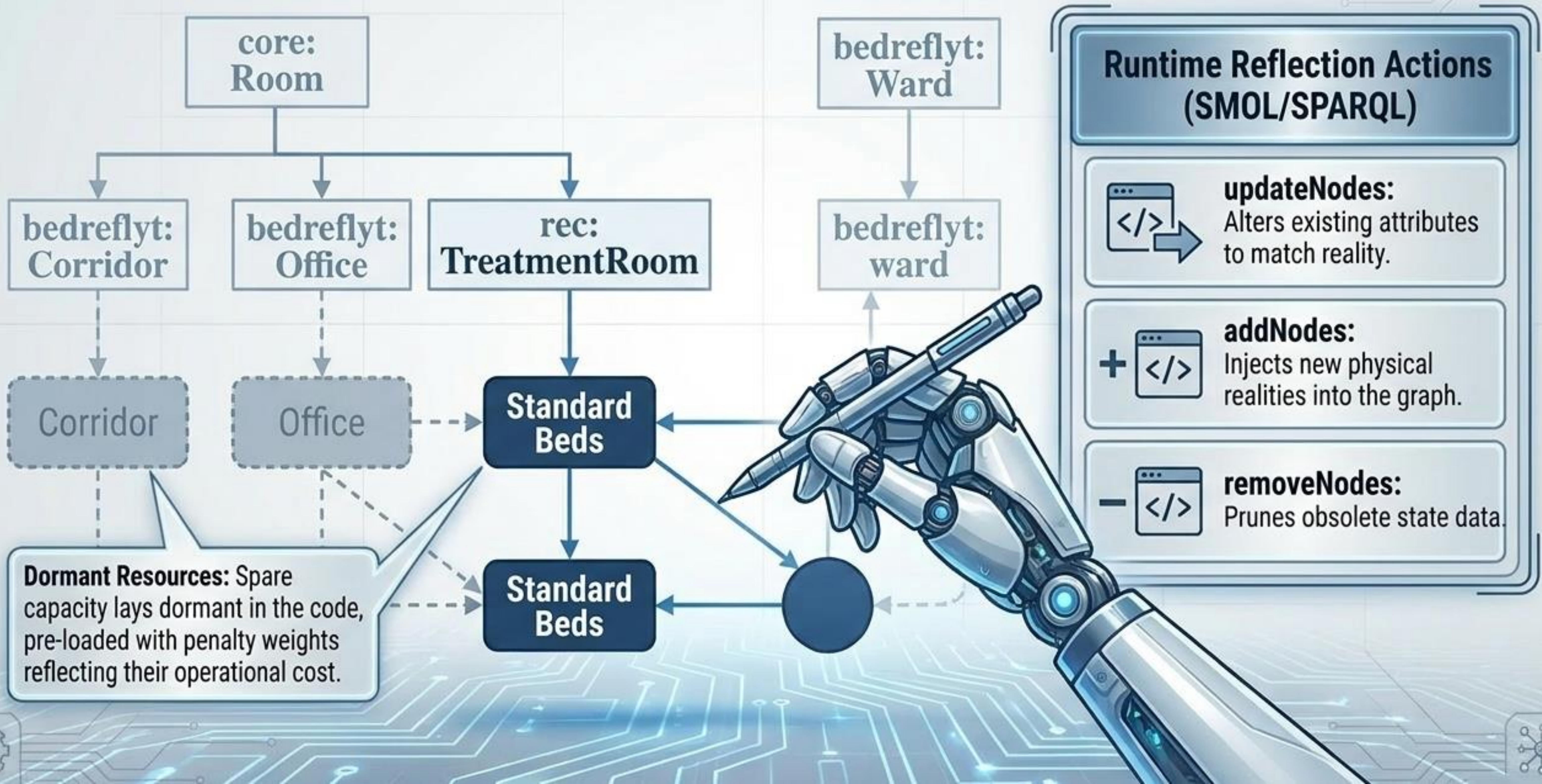
	Static Digital Twins	DynResDT (Self-Adaptive)
Architectural Structure	Fixed layouts and offline, manual model updates.	Runtime structural alignment via **Semantic Reflection.**
Demand Handling	Deadline-based, **rigid** operational policies.	Declarative **lifecycle-triggered adaptation.**
Crisis Decision Making	Ad-hoc, manual human-dependent tracking.	Penalty-guided, constraint-based mathematical optimization.

Goal: Systems that think, adapt, and optimize in real-time.

The DynResDT Architecture



Pillar I: The Evolving Semantic Model



Pillar II: Lifecycle State Management

Load State

Under-loaded if $0 \leq \Gamma \leq \Theta$

Over-loaded if $\Gamma > \Theta$

where $\Gamma = \lfloor 45/100 \cdot 90 \rfloor = 40$

The Formula

$$\Gamma = \left[\frac{\Omega \cdot \varepsilon}{100} \right]$$

The Threshold

ε is explicitly set to 90% of default capacity.

The Trigger

When occupied beds (Θ) cross the 90% threshold, the Lifecycle Manager instantly prompts the Orchestrator to search for a new optimal configuration.

Pillar III: Penalty-Guided Optimization

Powered by Optimization Modulo Theories (Z3)

$$\min \sum (\delta_r \cdot a_{id,r}) + \sigma_{id}$$

The Translation:
The algorithm mathematically minimizes the maximum penalty. It answers a crisis by deploying the least disruptive spare resources first—prioritizing low-penalty offices over high-contagion corridors, while minimizing patient movement (σ_{id}).

The Proving Ground: BedreFlyt Integration

Bed Bay Categories

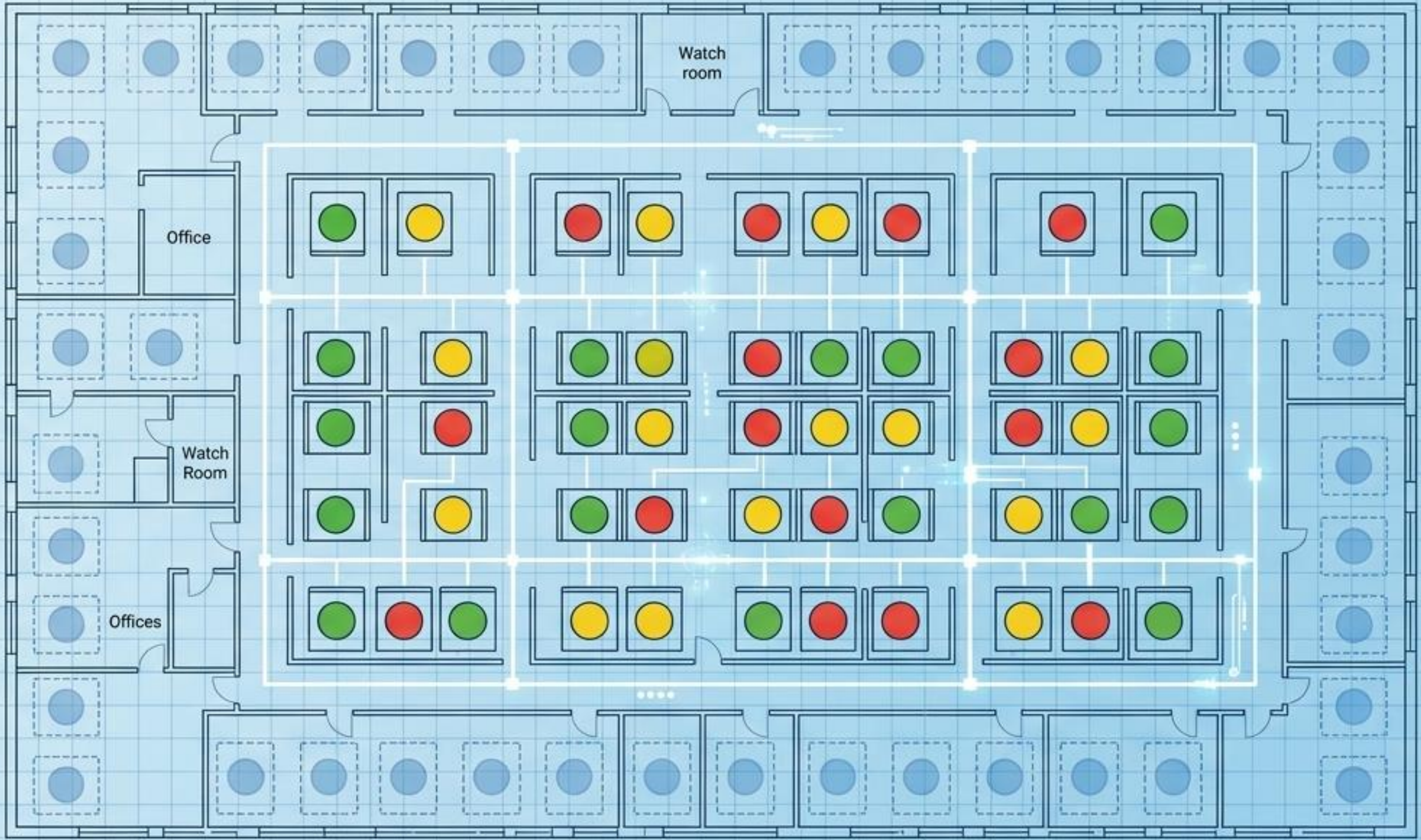
1 Standard · 2 Intermediate · 3 High Monitoring · Crisis

Simulation Goal

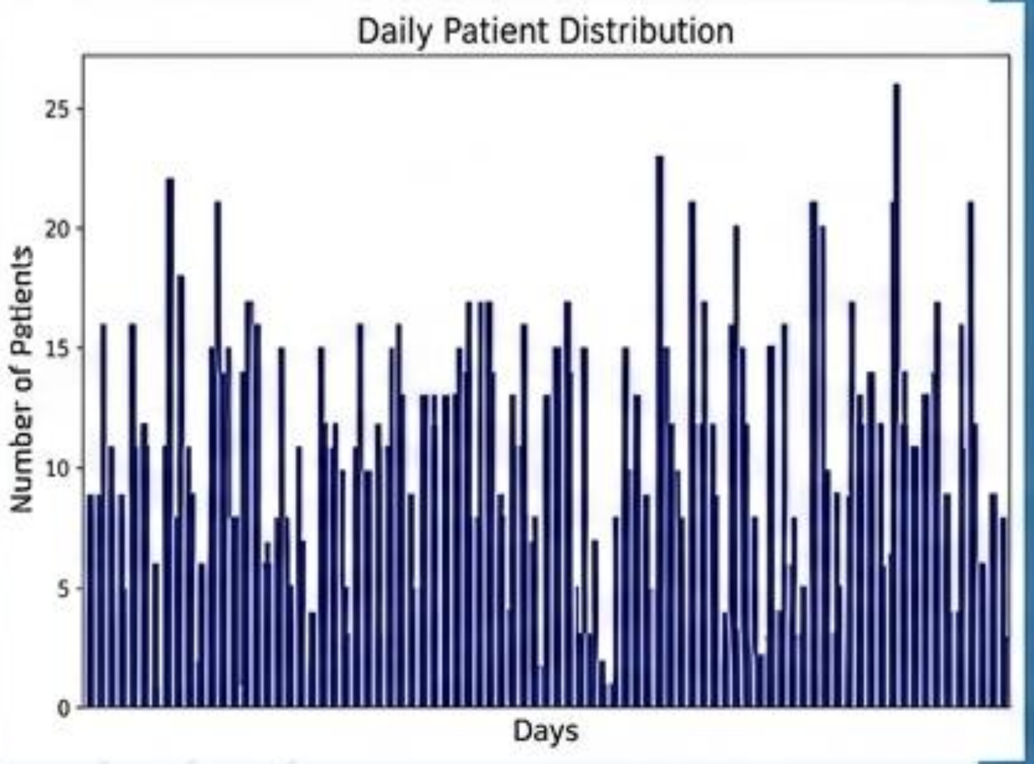
Transform a timed stream of incoming patients into a mathematically optimized Bed Bay Allocation stream.

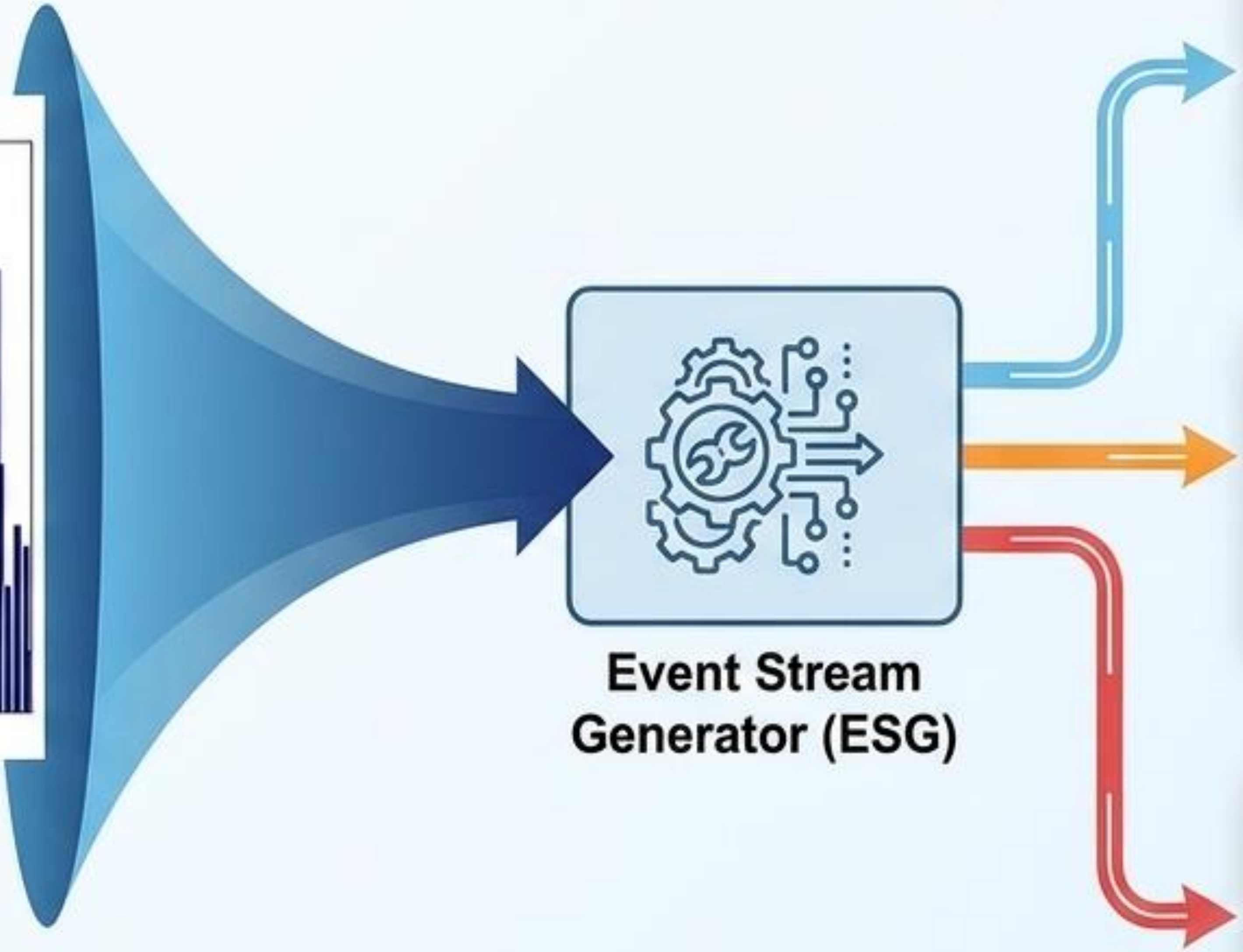
Scale Constraints

Default capacity is 45 beds. The system can autonomously scale up to a maximum of 93 beds if operational penalty thresholds are continuously breached.



Simulating Chaos: The Event Stream Generator





Baseline Data

Patient arrivals sampled from a Poisson distribution ($\lambda = 40$).

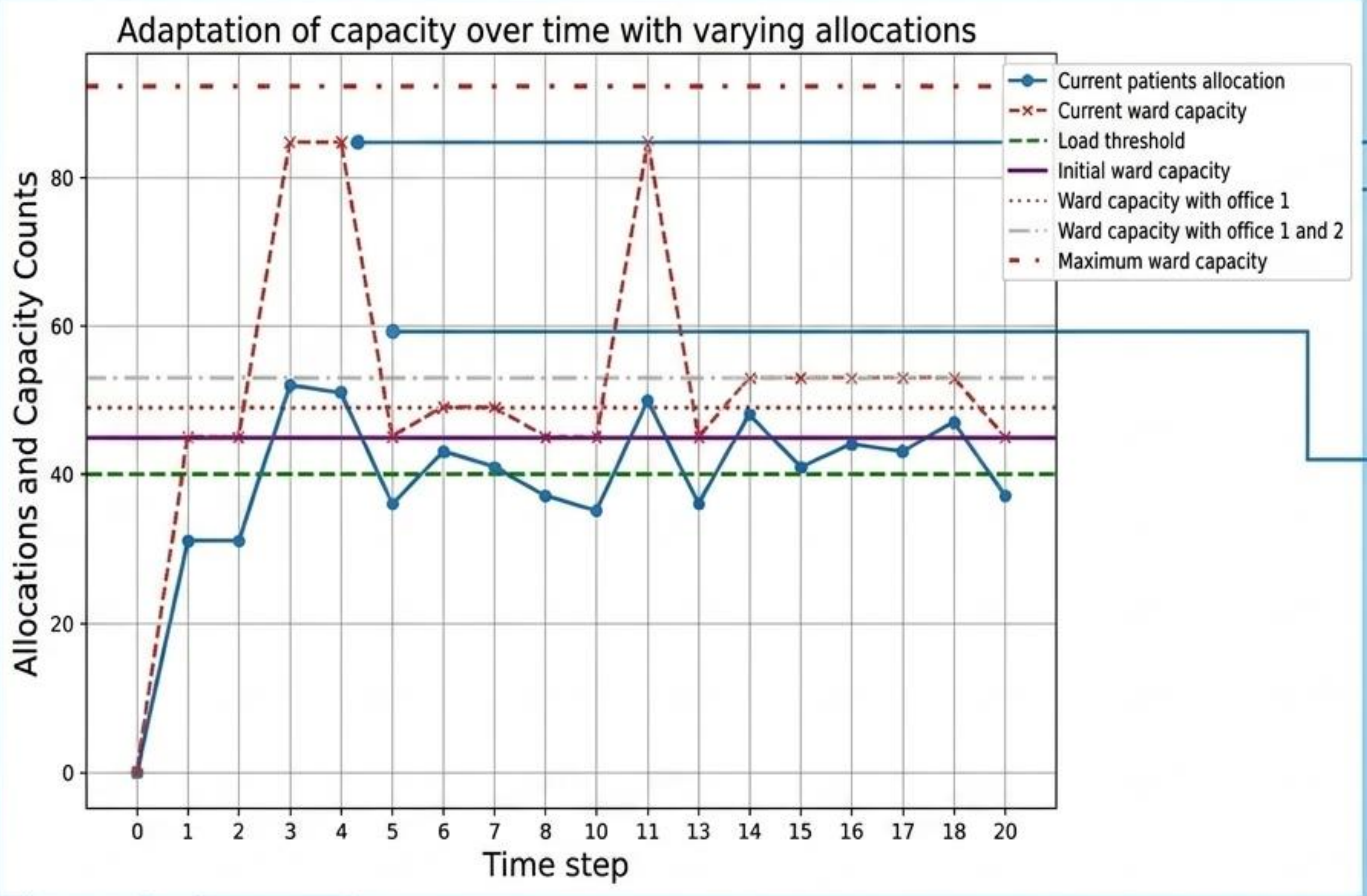
The Stress Test

Synthesizing 'Crisis Peaks'—switching loads every 5 time steps and multiplying occurrences by a massive 1.5 factor (ξ).

The Output

A modular, time-stamped stream of simulated patient arrivals mimicking an unpredictable, severe healthcare crisis.

The Stress Test: Measured Adaptation



The Breach

Patient load crosses the 90% threshold ($\Gamma = 40$). The Ward Capacity (red dashed line) steps up instantly.

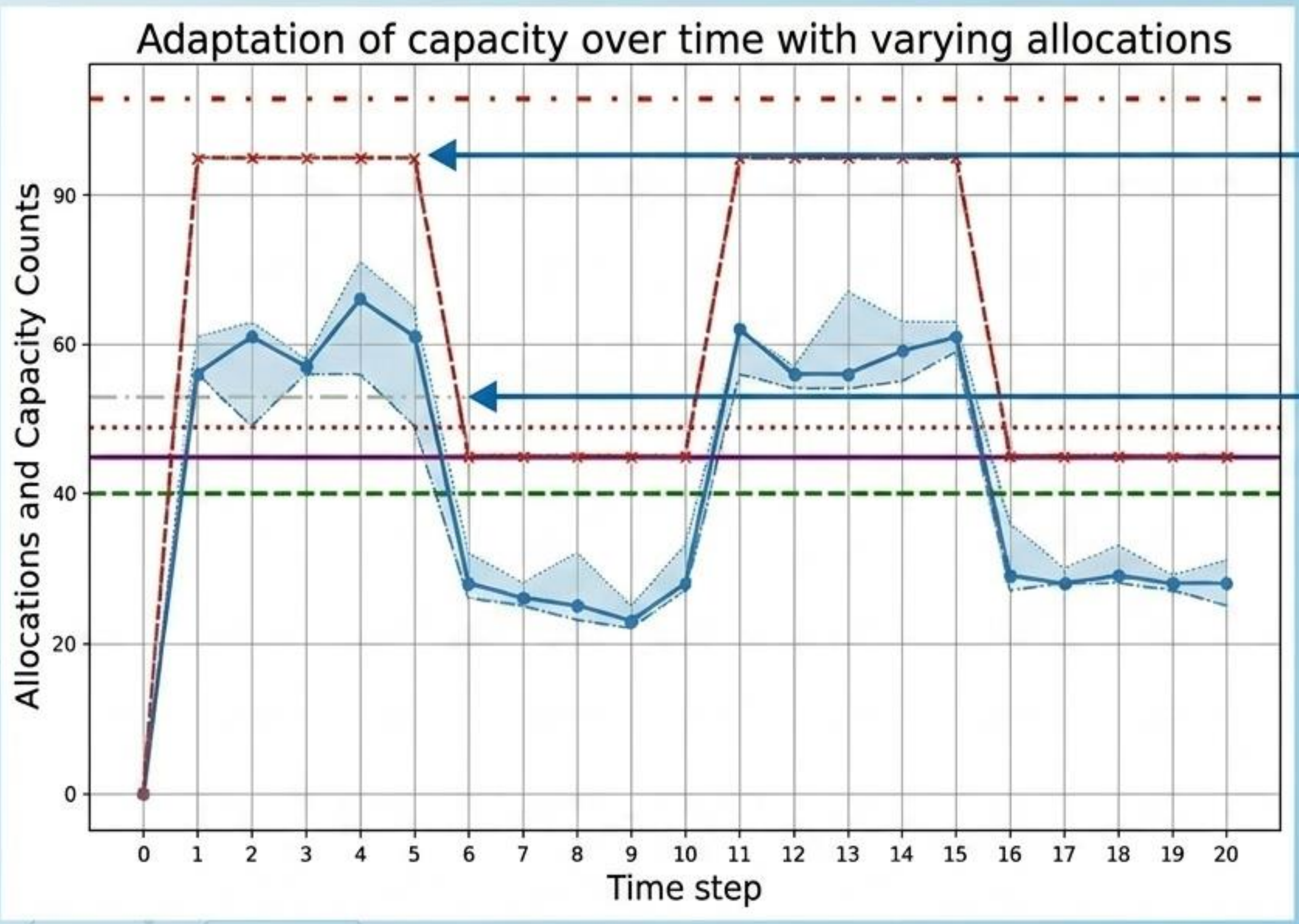
Precision in Action

At Time Step 5, a surge is handled by opening a single office. The system actively avoids over-reacting, skipping the high-penalty corridor.

Constraint Mandate

The system fulfills its mandate to minimize patient movement and only deploys the most costly resources as a last resort.

The Stress Test: Surviving Sustained Peaks



Macro-Resilience Over Time

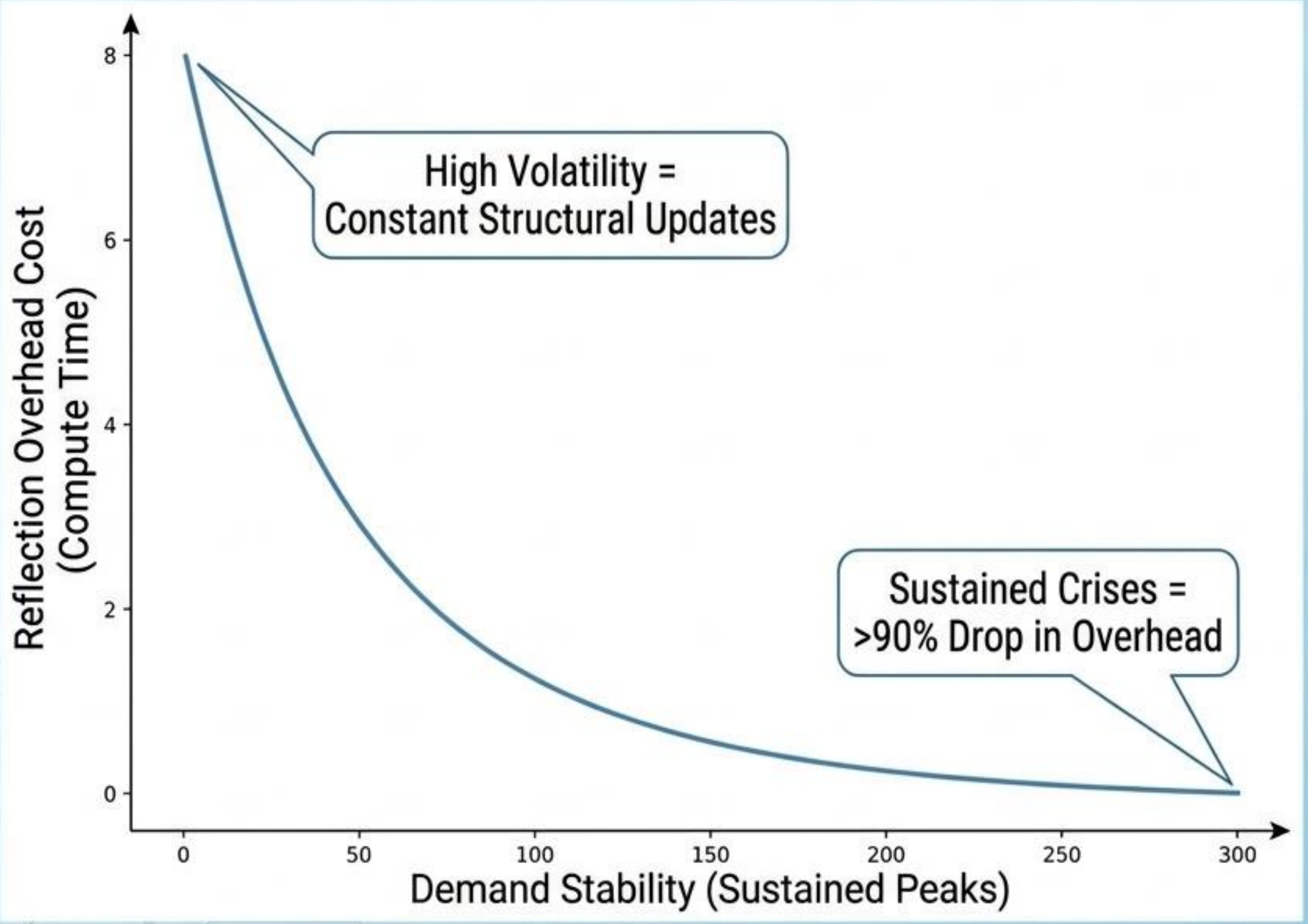
- **Elastic Expansion:** Notice the massive step-ups to maximum capacity (80+ beds) precisely when the crisis waves hit.
- **Algorithmic Retraction:** Crucially, observe the immediate step-downs when the surge subsides.
- **The Conclusion:** The Lifecycle Manager seamlessly opens overflow capacity to survive the peak, and closes it immediately to return to baseline efficiency and reduce penalties.

The Cost of Intelligence

High Fluctuation (Constant Structural Alignment)	Sustained Peak (Stable Overflow Capacity)
~286.8s	~21.3s
Average SMOL Alignment Overhead (Max: 3917s)	Average SMOL Alignment Overhead (Max: 474s)

The Architectural Trade-Off:
Self-adaptation is not free. Semantic reflection introduces measurable computational overhead. The architecture explicitly sacrifices micro-seconds of runtime efficiency to guarantee principled systemic correctness.

Architectural Lesson I: Stability vs. Cost



The Rule of Volatility:

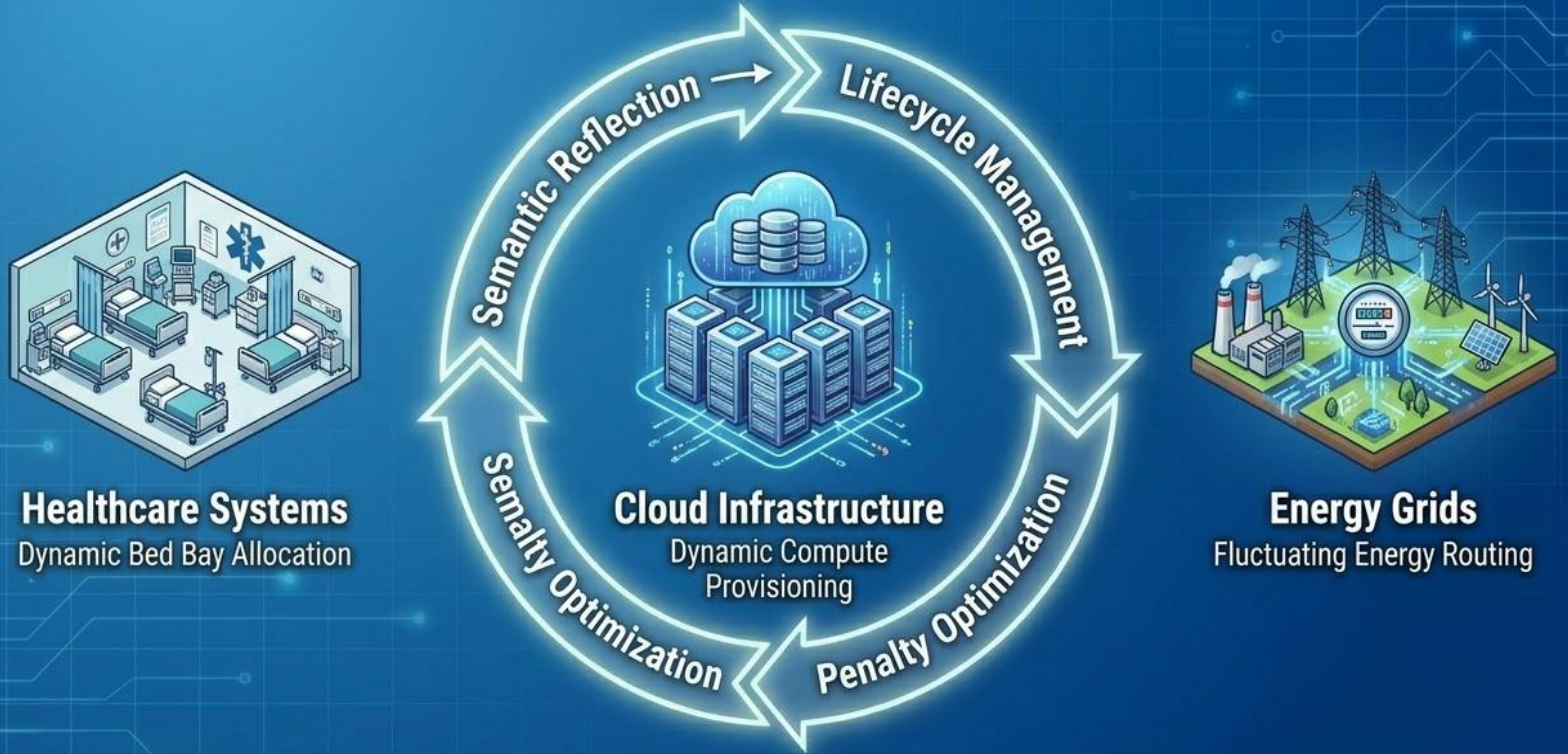
The computational cost of semantic reflection is directly proportional to environmental volatility, not system size.

During a sustained crisis where overflow capacity remains open, structural updates freeze. The system scale effortlessly via standard simulation techniques.

Architectural Lesson II: The Math of Resilience

Encoding Human Philosophy:
Penalties make the resilience-vs-cost relationship explicit. By encoding human operational ethics as quantitative mathematical penalties, the architecture guarantees that algorithmic decision-making aligns with human values. The system only sacrifices optimal conditions when mathematically compelled to by the crisis.

A Universal Blueprint



While validated in high-stress hospital simulations, **DynResDT** provides a **domain-agnostic blueprint**. Any environment where physical constraints fluctuate unpredictably can utilize this self-adaptive triad to survive the peak.

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